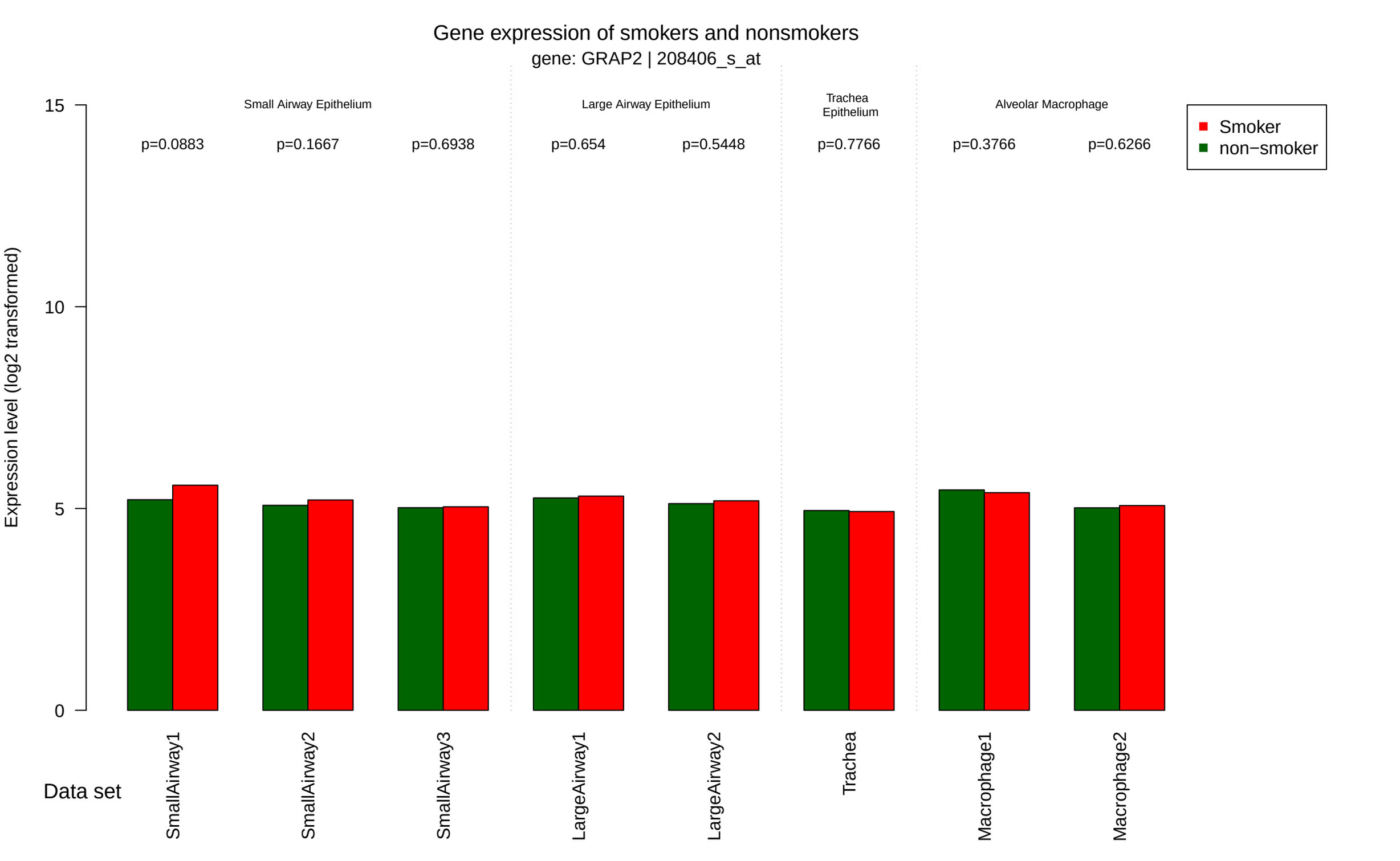
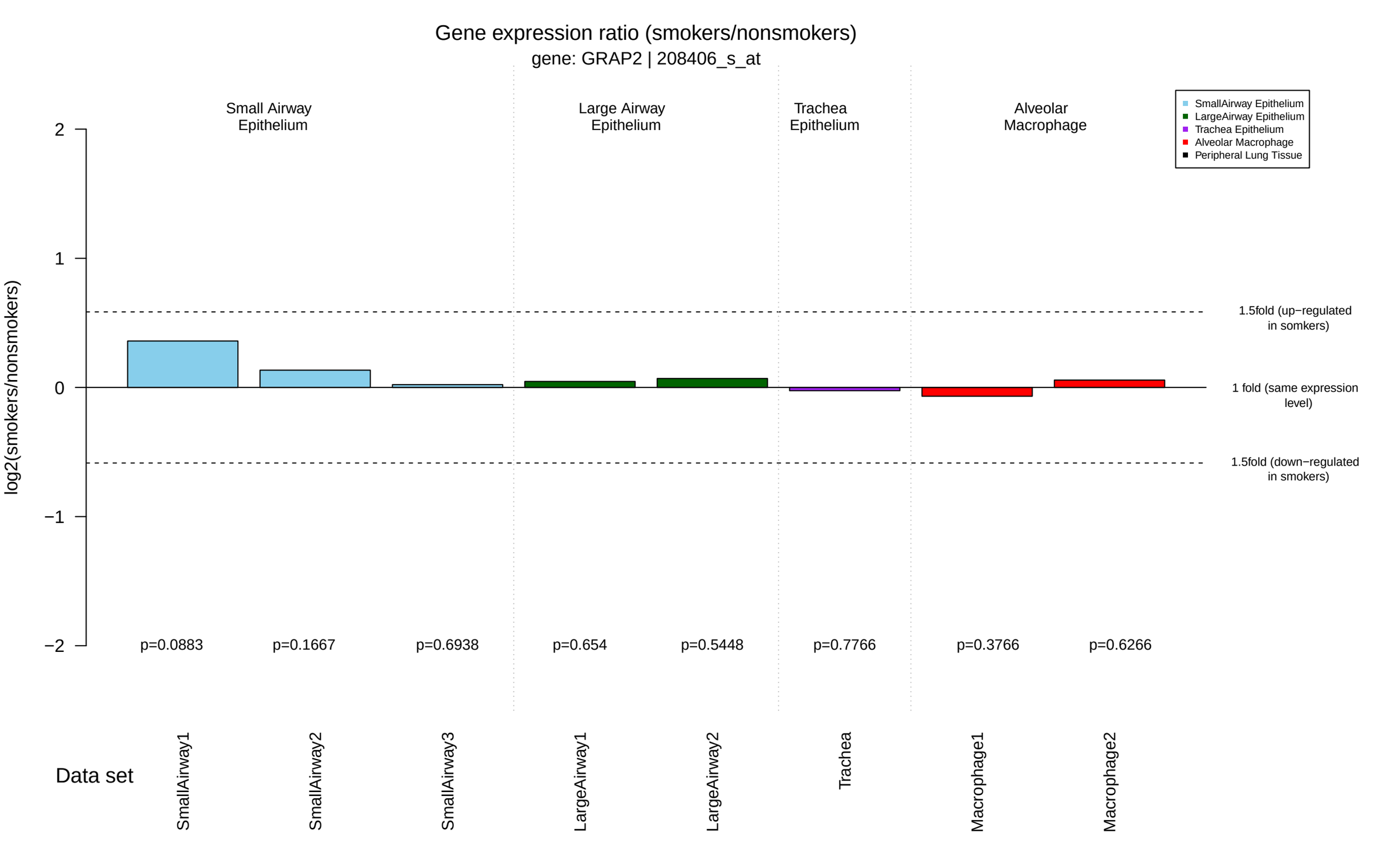
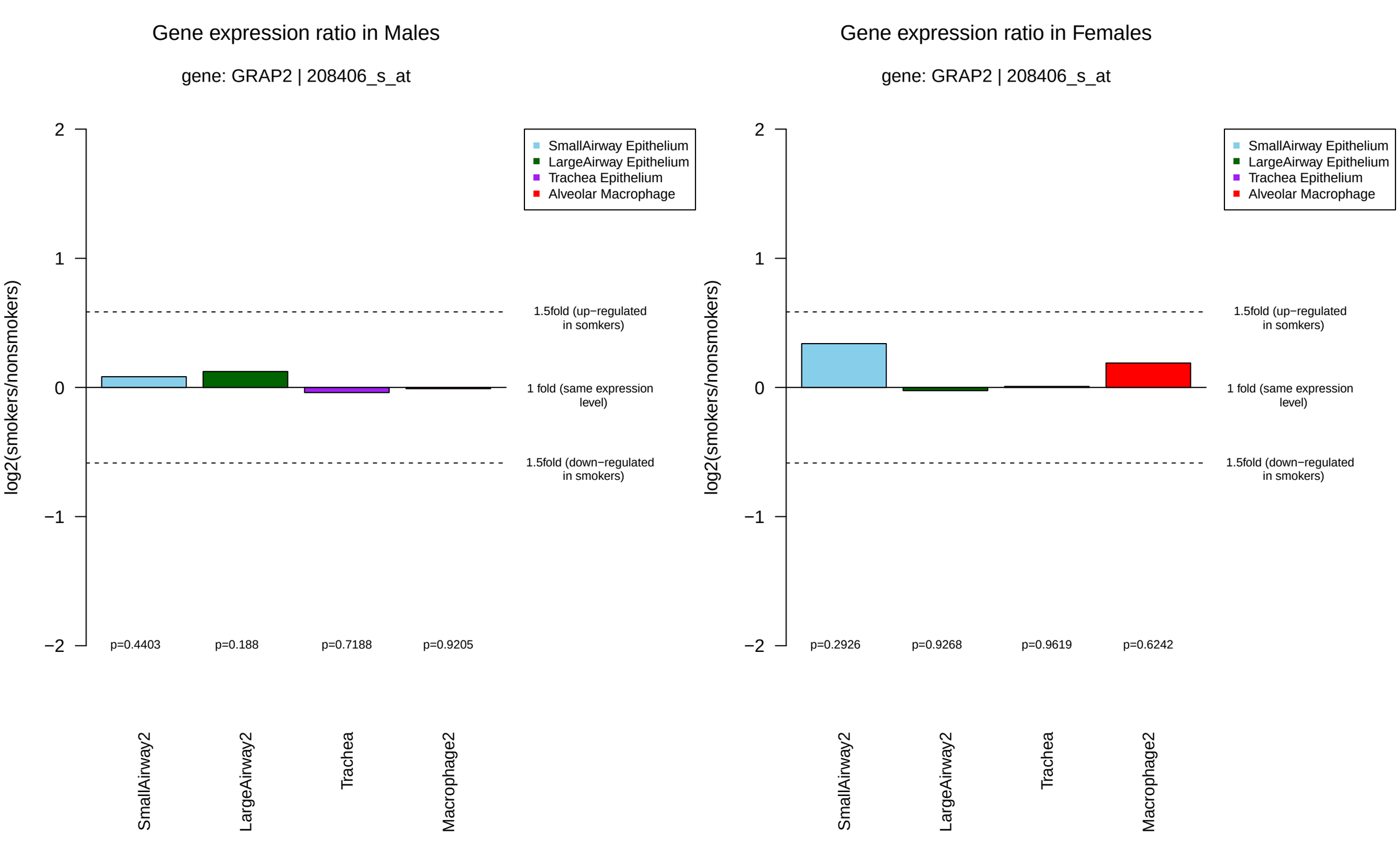
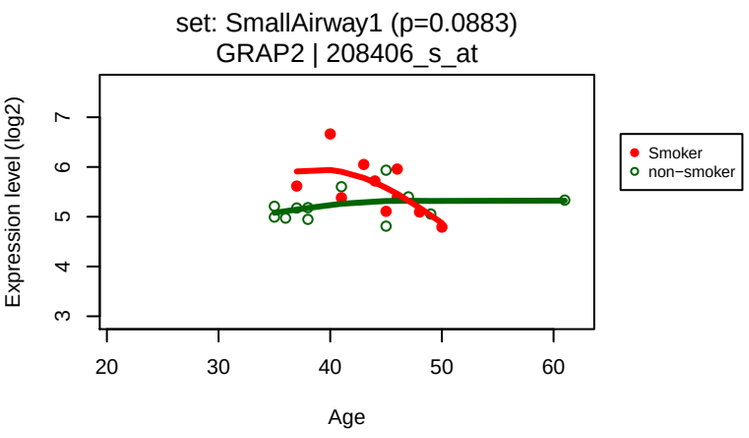
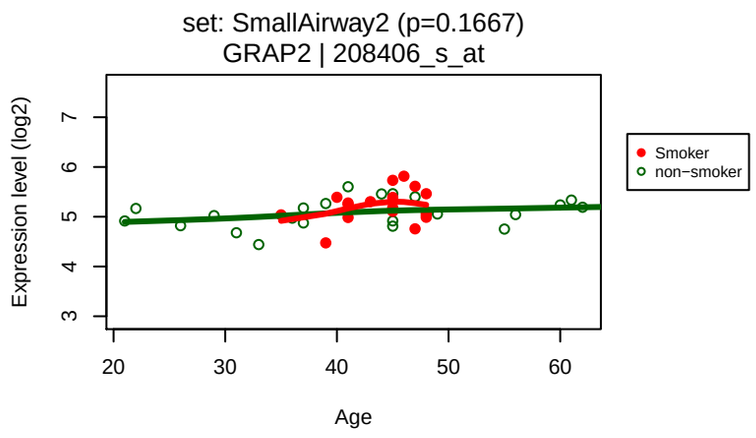
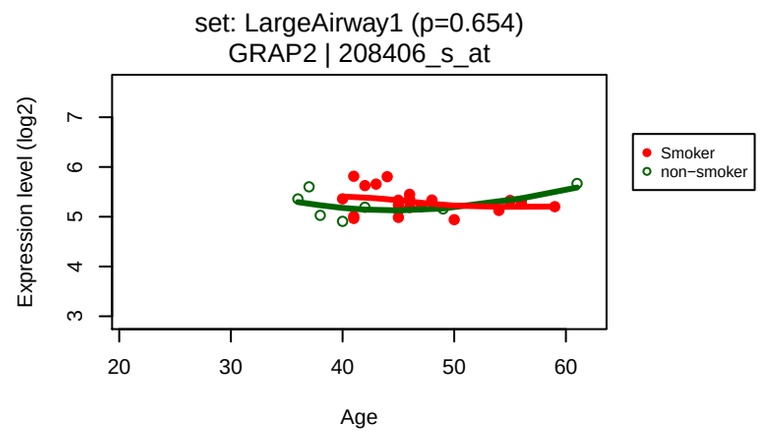

Gene expression of smokers and nonsmokers
gene: GRAP2 | 208406_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0883
p=0.1667
p=0.6938
p=0.654
p=0.5448
p=0.7766
p=0.3766
p=0.6266
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

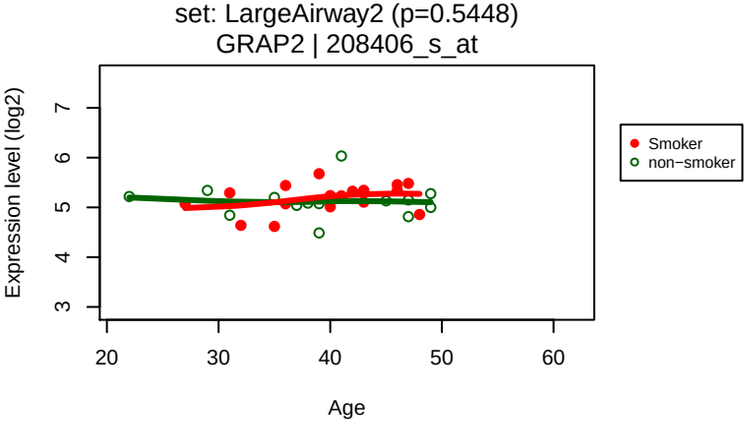

Gene expression ratio (smokers/nonsmokers)
gene: GRAP2 | 208406_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0883
p=0.1667
p=0.6938
p=0.654
p=0.5448
p=0.7766
p=0.3766
p=0.6266
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

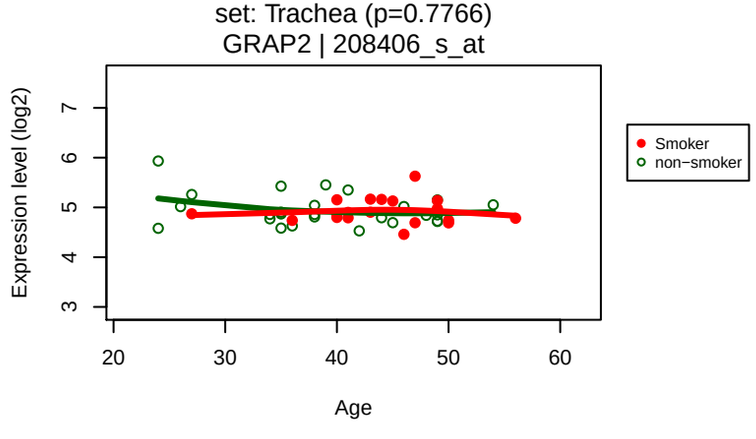

Gene expression ratio in Males
gene: GRAP2 | 208406_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4403
p=0.188
p=0.7188
p=0.9205
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRAP2 | 208406_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2926
p=0.9268
p=0.9619
p=0.6242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

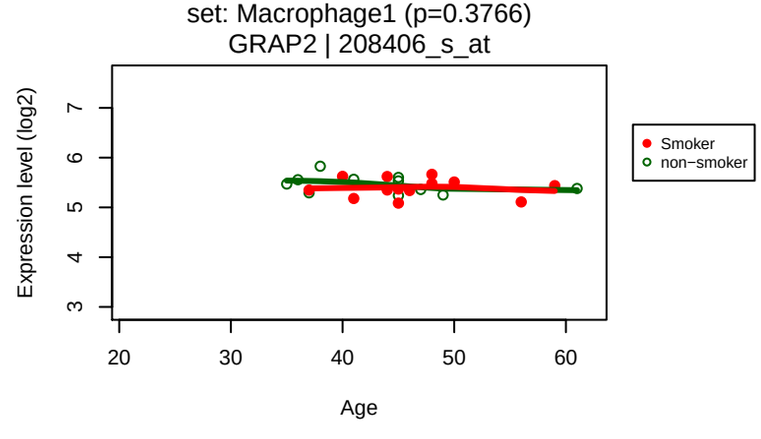

set: SmallAirway1 (p=0.0883)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker

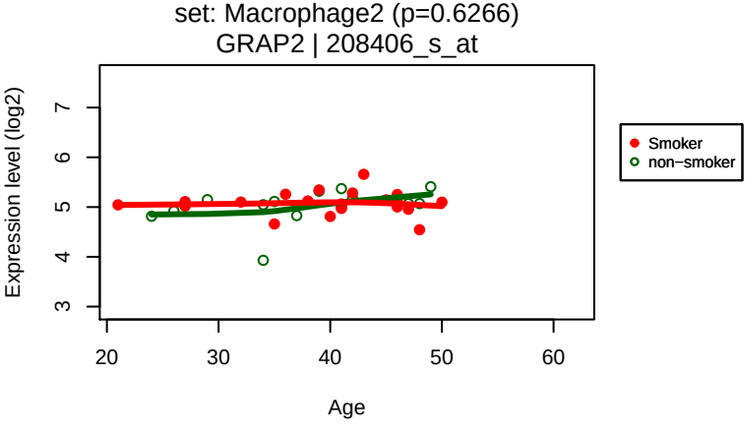

set: SmallAirway2 (p=0.1667)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker

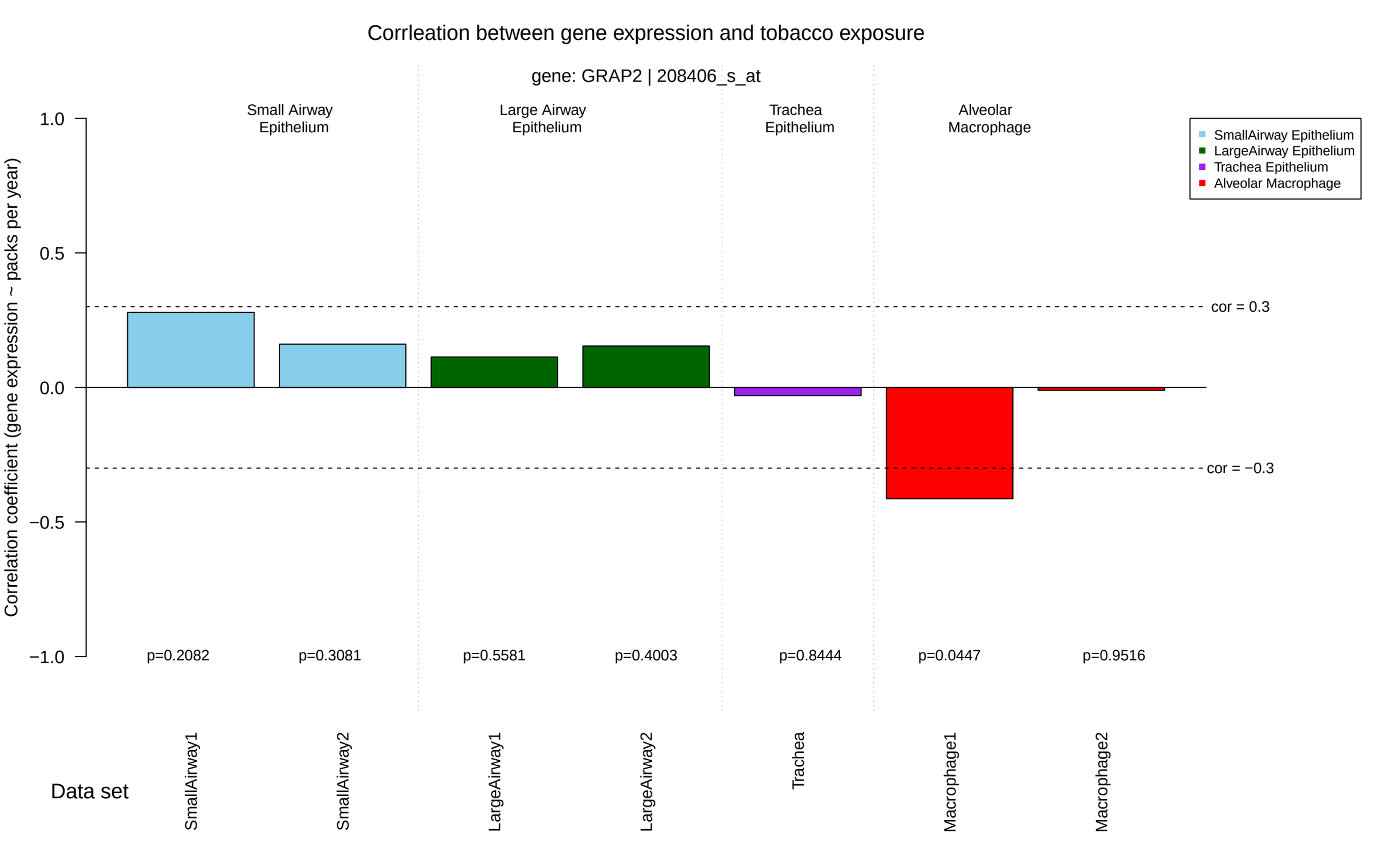

set: LargeAirway1 (p=0.654)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5448)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7766)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3766)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6266)
GRAP2 | 208406_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRAP2 | 208406_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2082
p=0.3081
p=0.5581
p=0.4003
p=0.8444
p=0.0447
p=0.9516
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2